

RedScope Capstone Project

Lab-only Red-Team Assessment with Detection and Remediation Focus

Shamanth R Singh

Network foothold and post-exploitation

Ajit Mohan

Scope, web exploit summary, coordination

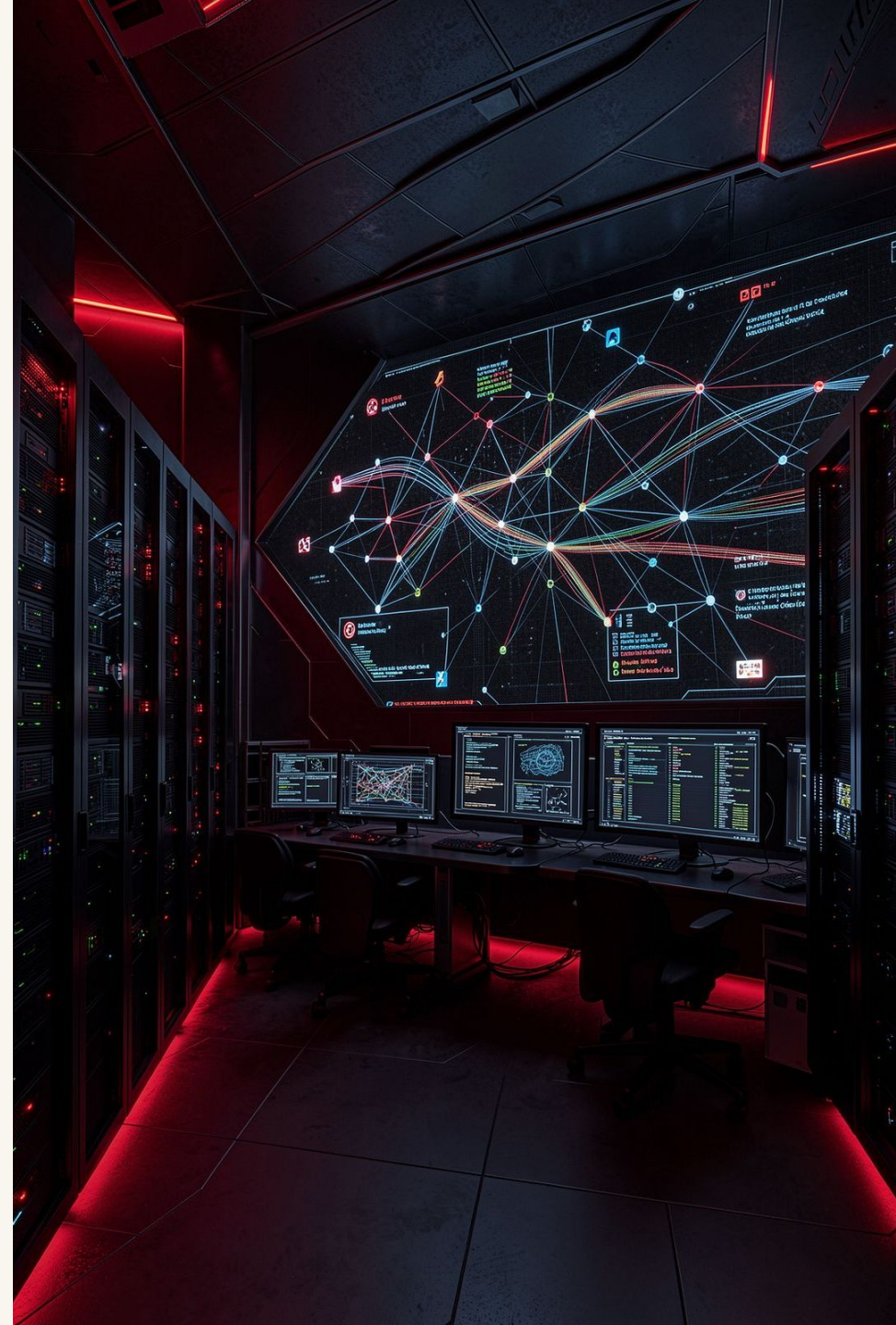
Saiteja Kacham

ML detector and evasion work

Madhurjya Deka

Remediation and business impact

Speaker: Ajit Mohan





Project Objective and Why It Matters

Objective

- Simulate a realistic attacker path in a safe lab environment
- Identify how web and network weaknesses can lead to broader compromise
- Test detection logic, including ML-based detection
- Translate technical findings into business risk and remediation priorities

Why It Matters

- Security gaps rarely exist in isolation
- Attackers chain small weaknesses
- Detection must be validated, not assumed
- Remediation must be risk-based and business-aligned

Speaker: Ajit Mohan

Scope and Rules of Engagement

In Scope

- Lab-only environment — controlled assets and test systems
- Web exploitation assessment
- Network compromise simulation
- Post-exploitation observation
- ML detector and evasion testing
- Remediation and risk mapping

Out of Scope

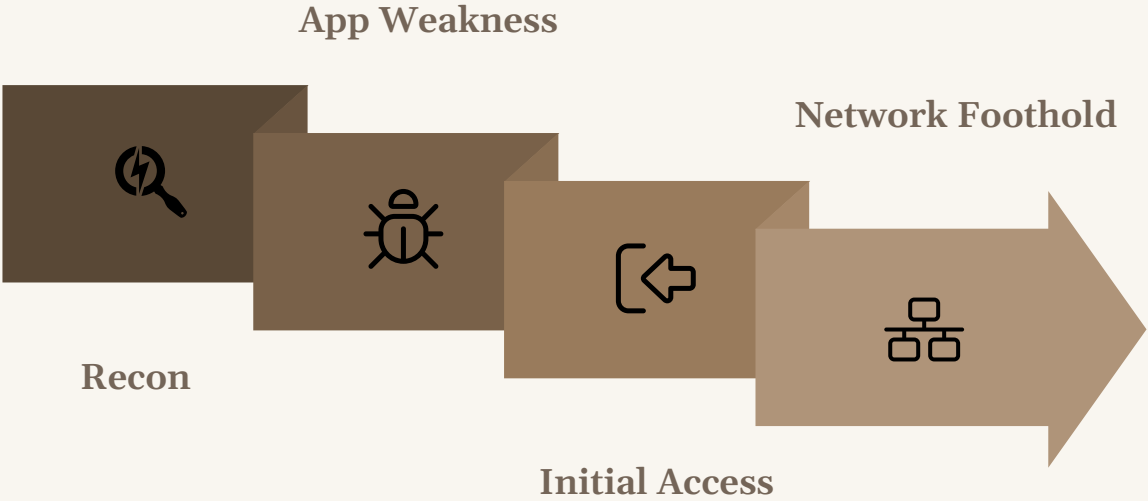
- Real public systems
- Unauthorised access
- Credential abuse outside lab
- Destructive payloads
- Persistence beyond test intent

📌 All activity was conducted in a controlled, isolated lab environment only.

Speaker: Ajit Mohan

End-to-End Attack Chain Overview

RedScope Assessment Flow — eight sequential phases from reconnaissance to business impact mapping.



Each phase builds on the previous, demonstrating how a single weakness can chain into a broader compromise — and where defenders can intervene at every stage.

Handoff: Ajit Mohan to Shamanth R

Web Exploit Summary

Web Exploitation Focus

Surface Identification

Identify exposed web attack surface across the lab environment

Input and Auth Analysis

Analyse input handling and authentication-related weaknesses

Chaining Validation

Validate whether weaknesses can be chained for greater impact

Documentation

Document findings with severity and reproducibility ratings

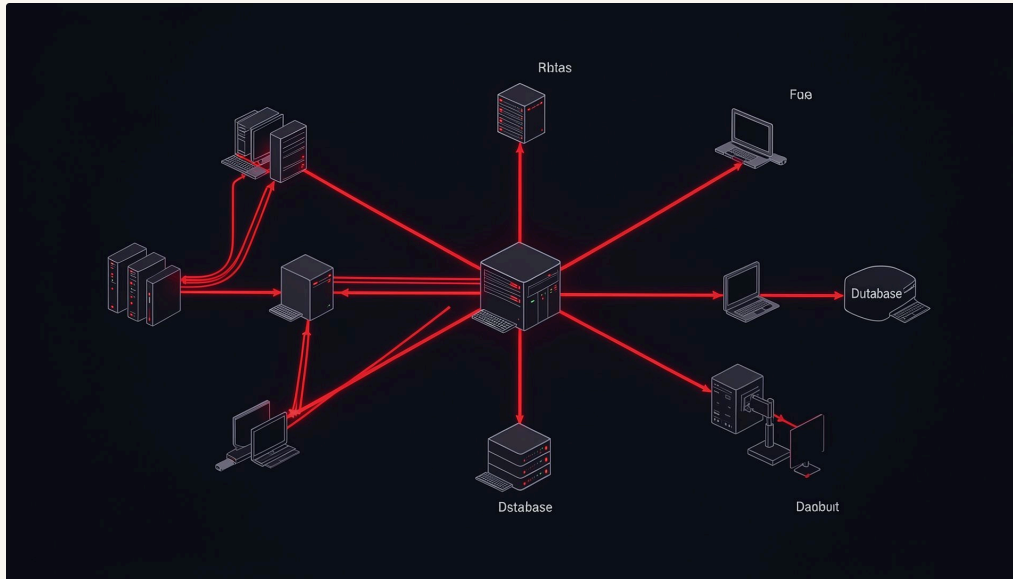
Defensive Lessons

Capture defensive lessons from the web layer

Key Learning: Web weaknesses become more serious when they enable movement into deeper systems.

Speaker: Shamanth R Singh

Transition: Web Layer to Network Context



From Web Layer to Network Context

→ Initial Weakness Creates Opportunity

A web-layer finding opens the door to the internal network

→ Network Exposure Determines Blast Radius

Reachable systems define how far an attacker can move

→ Privilege Boundaries Become Critical

Least privilege limits what an attacker can do post-access

→ Monitoring Must Capture Lateral Movement

Detection must extend beyond the web perimeter

→ Segmentation Reduces Attacker Options

Proper segmentation contains the impact of initial compromise

Speaker: Shamanth R Singh



Network Foothold and Post-Exploitation

Network Foothold Analysis

- Understand what access is possible after initial compromise
- Identify reachable systems and exposed services
- Evaluate segmentation and privilege boundaries
- Observe attacker-like behaviour in a controlled lab
- Map potential paths to sensitive assets

Post-Exploitation View

What could an attacker see?

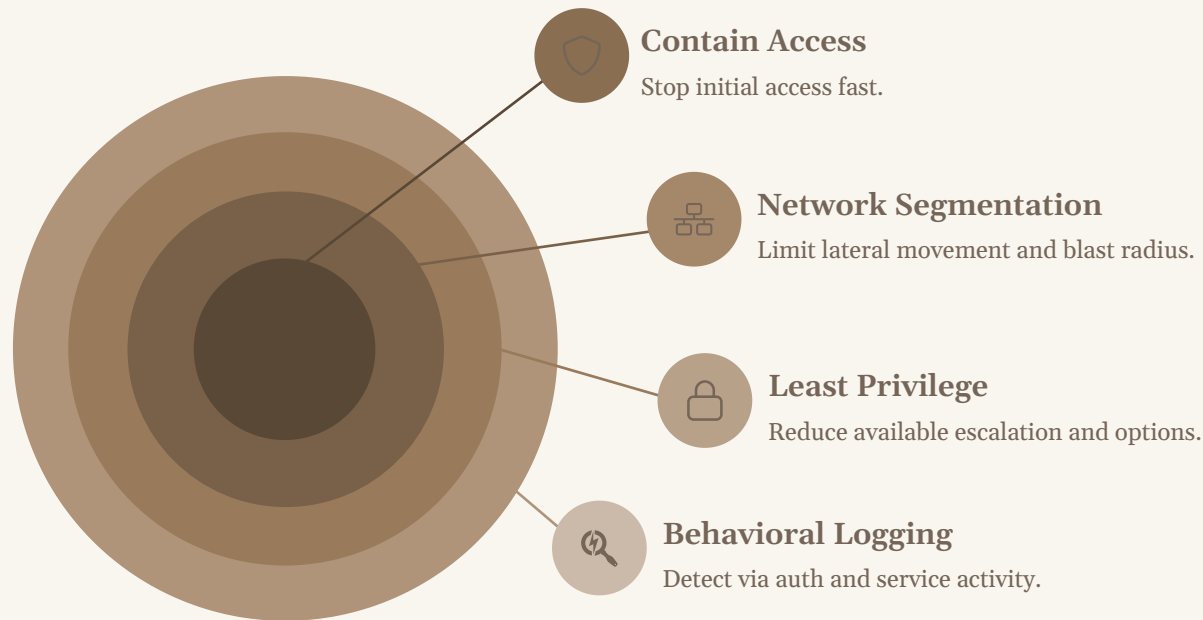
What could they access?

What traces would defenders need?

Which controls limit impact?

Speaker: Shamanth R Singh

Post-Exploitation Findings and Defensive Lessons



Defensive Lessons

- Initial access must be contained quickly
- Network segmentation limits blast radius
- Least privilege reduces movement options
- Logging must cover authentication, service access, and unusual behaviour
- Detection should focus on behaviour, not only signatures

Security Principle: Assume compromise. Design controls to limit movement and detect abnormal activity.

Handoff: Shamanth R to Saiteja Kacham



ML Detector: Purpose and Approach

ML Detection Objective

- Detect suspicious or attacker-like behaviour
- Learn from patterns in security-related data
- Classify activity as benign or suspicious
- Evaluate detector performance
- Understand limitations under adversarial conditions

Why ML?



Pattern Recognition

Can identify patterns beyond static rules



High-Volume Analysis

Helps analyse high-volume signals efficiently



Careful Validation

Requires careful validation and tuning

Speaker: Saiteja Kacham

ML Evasion and Detection Limitations



Evasion Testing Focus

- Understand how attackers may alter behaviour to avoid detection
- Test whether small changes reduce detection confidence
- Identify blind spots in the detector
- Improve model robustness through adversarial testing
- Recommend layered detection strategies



Important Point: ML should complement security controls, not replace them.

Speaker: Saiteja Kacham

Risk Summary: Technical Findings to Business Impact

Each technical finding maps directly to a business risk. Understanding this chain is essential for prioritising remediation investment.

Technical Finding	Business Risk	Impact Level
Web weakness	Unauthorised access and data exposure	High
Network foothold	Lateral movement and wider compromise	High
Weak segmentation	Larger blast radius across systems	High
Limited logging	Delayed detection and investigation	Medium
ML evasion possibility	False negatives and missed alerts	Medium

Speaker: Madhurjya Deka

Remediation Strategy



Speaker: Madhurjya Deka

Recommended Remediation Priorities

01

Fix Web Application Vulnerabilities

Address identified weaknesses at the web layer first

02

Strengthen Authentication and Session Controls

Harden login flows and session management

03

Apply Least Privilege Across Systems

Restrict access rights to the minimum required

04

Enforce Network Segmentation

Isolate systems to contain potential compromise

05

Improve Logging, Alerting, and ML Validation

Harden exposed services, expand coverage, and test detection adversarially

06

Build an Incident Response Playbook

Prepare structured response procedures for identified scenarios

Business Impact and Value of RedScope

Business Impact

- Reduces likelihood of unauthorised access
- Limits blast radius after initial compromise
- Improves detection and response readiness
- Helps prioritise remediation investment
- Supports security awareness across technical and business teams

Offence

Realistic attack simulation

Detection

ML-aware validation

Remediation

Business-aligned priorities

Value of the Project

RedScope connects offence, detection, and remediation into one practical security lifecycle.

Speaker: Madhurjya Deka



Key Takeaways

1 Safe, Lab-Only Assessment

RedScope is a controlled, integrated security assessment conducted entirely within a lab environment

2 Web Weaknesses as Entry Points

Web vulnerabilities can become the starting point for broader, chained compromise

3 Segmentation and Least Privilege Are Critical

These two controls are the most effective at limiting attacker movement post-access

4 ML Detection Must Be Validated

ML is useful but must be tested against evasion to understand its real-world limits

5 Remediation by Business Impact

Prioritise fixes based on the business risk each finding represents

6 Prevention + Detection + Response

Security is strongest when all three pillars work together in an integrated lifecycle

Speaker: Shamanth R Singh

Thank You

Questions?

Suggested Q&A Areas

Lab-Only Rationale

Why was the project conducted in a lab-only environment?

Scope Control

How was scope defined and controlled throughout the assessment?

ML Inclusion

Why include ML detection in a red-team assessment?

Remediation Priorities

What were the main remediation priorities identified?

Enterprise Applicability

How can this model apply to real enterprise environments?

Speaker: Shamanth R Singh — with all team members answering their respective sections

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